higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

BUILDING AND STRUCTURAL CONSTRUCTION N5

(8060015)

2 December 2019 (X-Paper)
09:00–13:00

REQUIREMENTS: **Answer book (8/13)**
A2 drawing sheets
Hot-rolled steel sections (BOE 8/2)

Nonprogrammable calculators may be used.

This question paper consists of 5 pages, 2 diagram sheets, 1 formula sheet and 2 addenda.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
BUILDING AND STRUCTURAL CONSTRUCTION N5
TIME: 4 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.

2. Read ALL the questions carefully.

3. QUESTIONS 1, 2 and 6 must be done in the ANSWER BOOK.
   QUESTIONS 3 and 4 must be done on the drawing paper (supplied).
   QUESTION 5 must be done on ADDENDUM 1 (attached).

4. Number the answers according to the numbering system used in this question paper.

5. ALL drawings must be done in pencil with bold outlines.

6. The drawings must be done in accordance to the National Standards and be fully labelled with descriptive notes and dimensions (where applicable).

7. ALL calculations must conform to the relevant SABS/SANS Code of Practices.

8. Write neatly and legibly.

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QUESTION 1

DIAGRAM SHEET 1, FIGURE 1 (attached) shows a U-shaped figure supporting a rectangular steel plate.

Calculate the following:

1.1 The distance of the neutral axis about A–B (5)

1.2 The second moment of area for the given figure (7)

1.3 The profile modulus about the x–x axis (2)

1.4 The maximum bending moment for the given section. Use a bending stress of 145 MPa. (2)

[16]

QUESTION 2

DIAGRAM SHEET 1, FIGURE 2 (attached) shows a steel beam supported on TWO reactions, 9 metres apart, then extends a further 2 metres to form a balcony. The steel beam will be required to carry THREE point loads and TWO uniformly distributed loads.

Select a suitable I-section parallel flange steel beam to ensure the building is safe.

Use the following scales:

Load diagram: 140 mm = 11 m
Shear force: 2 mm = 1 kN
Bending moment: 3 mm = 1 kN.m
The permissible bending stress for grade 43 mild steel is 160 MPa.
The maximum shear stress must not exceed 100 MPa.

2.1 Calculate the value of the reactions at RL and RR. (4)

2.2 Determine the values of the shear forces at each intersection. Draw the shear force diagram. Insert all the values on the diagram. (4)

2.3 Calculate and draw the bending moment diagram. Insert the value of the bending moment's maximum in the diagram. (6)

2.4 Calculate the maximum section modulus for the loaded beam. (3)

2.5 Select a suitable I-section parallel flange for the loaded beam from the BOE 8/2 steel tables. (1)

2.6 Calculate the maximum shear stress in the chosen steel beam which should not exceed 100 MPa. (3)

[21]

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QUESTION 3

A 203 × 203 × 46,2 kg/m universal column is required to support TWO 356 × 171 × 44,8 kg/m main beams, welded in the centre of the flanges of the universal column. A 254 × 152 × 59,5 kg/m taper flange beam is welded centrally to the web of the column. The top of the beams must be on the same level.

Use scale 1:5 and draw the following views in first-angle orthographic projection.

3.1 The front view showing the taper flange beam in section

3.2 The side view showing the I-parallel flange in section

3.3 The top view of the steel column construction

The completed drawing must be drawn in first-angle orthographic projection and ALL the labels and hatching symbols must be included. **[20]**

QUESTION 4

Use scale 1:10 and draw the bottom FOUR steps of a reinforced concrete staircase and ground floor. FIGURE 3 below shows the layout of the steps. The staircase is anchored to a 300 × 450 mm deep edge beam, which also forms part of the first step.

The waist of the staircase is 170 mm thick and the reinforced concrete floor 100 mm thick. The floor and the steps of the staircase will receive a 19 mm cement screed finish while the underside (soffit) of the waist will be plastered.

REINFORCEMENT SPECIFICATIONS

Edge beam:	2Y16 main steel, 2Y12 compression steel and R10 binders Y10 starter bars
Staircase waist:	Y12 main steel at 200 mm centres and Y10 distribution steel at 200 centres
Floor reinforcement:	Y12 main steel and distribution steel at 200 centres, both ways

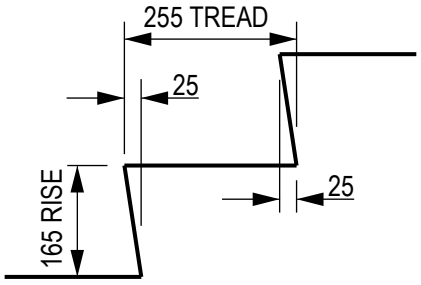


FIGURE 3

The completed drawing must include ALL the labels, dimensions and correct positioning of the reinforcement. **[20]**

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QUESTION 5

DIAGRAM SHEET 2, FIGURE 4 (attached) shows a longitudinal sectional view and sectional end view of a reinforced concrete beam to be cast over a shopfront window.

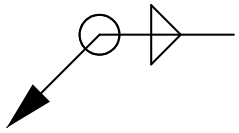
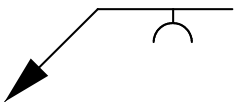
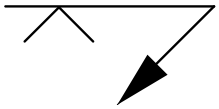
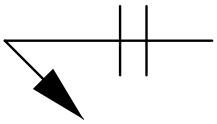
Remove ADDENDUM 1 (attached), print your EXAMINATION NUMBER in the provided space and complete the bending schedule TABLE for TWO such beams.
Use the shape code schedule on ADDENDUM 2 (attached).

Allow 150 mm for the bent-up bars and 100 mm for the binders.

NOTE: Do not write the same bar mark twice. **[15]**

QUESTION 6

Match a name and description of a welding symbol in COLUMN B with a welding symbol in COLUMN A. Write only the letter (A–F) next to the question number (6.1–6.4) in the ANSWER BOOK.

COLUMN A WELDING SYMBOLS		COLUMN B NAMES AND DESCRIPTIONS	
6.1		A	V-weld – on the arrow side
		B	J-weld – other side
		C	Square weld – both sides
6.2		D	Bevel weld – other side
		E	Fillet weld – both sides and all round
6.3		F	U-weld – on the arrow side
6.4			

(4 × 2) **[8]**

TOTAL: 100

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DIAGRAM SHEET 1

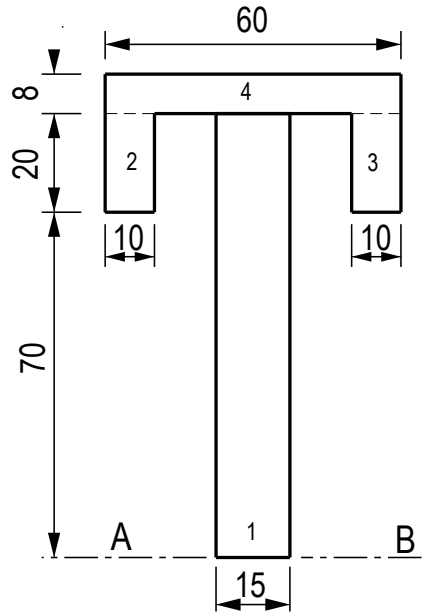


FIGURE 1

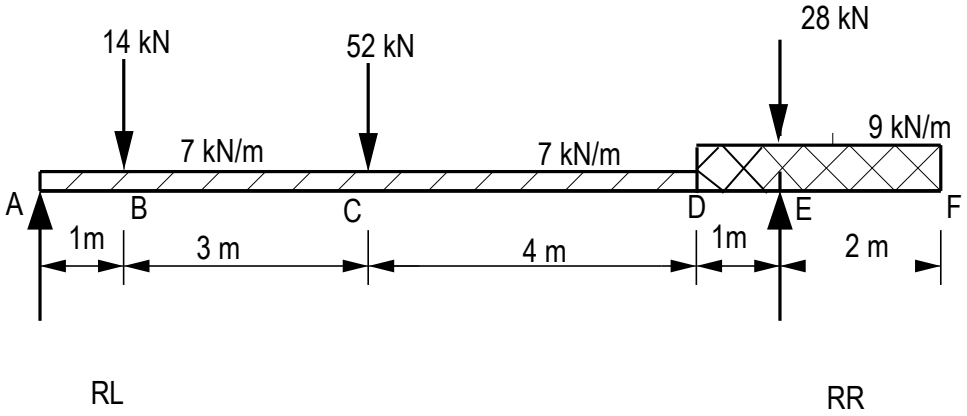


FIGURE 2

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DIAGRAM SHEET 2

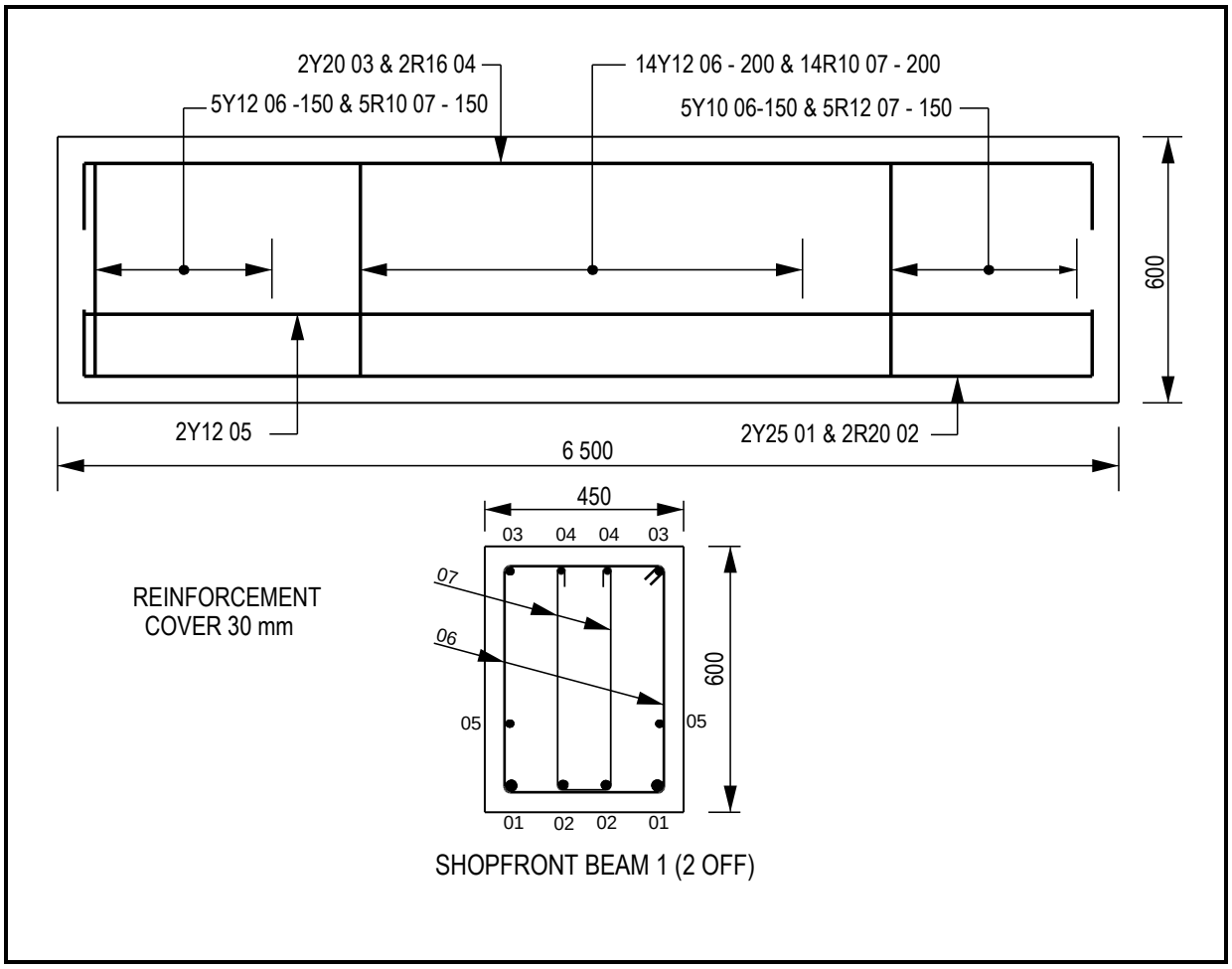


FIGURE 4

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BUILDING AND STRUCTURAL CONSTRUCTION N5

FORMULA SHEET

Any other applicable formula may also be used.

$A = \frac{1}{2}\ bh$

$A = \frac{\pi d^2}{4}$

$\sum CWM = \sum ACWM$

$\sum \uparrow F = \sum \downarrow F$

$Moment = Force\ x\ distance$

$BM = Reaction\ moment - Load\ moment$

$\frac{m}{I} = \frac{f}{y}$

$Z = \frac{I}{y}$

$I_{xx} = \frac{bd^3}{12}$

$I_{xx} = \frac{bd^3}{36}$

$I_{xx} = \frac{bd^3}{12} + al^2$

$F_s = A_s\ x\ f_s\ x\ n$

$F_t = A_t\ x\ f_t\ x\ n$

$F_c = A_c\ x\ f_c\ x\ n$

$A_s = \frac{\pi d^2}{4}$ $A_s = \frac{\pi(\emptyset - 0{,}9382\ p)^2}{4}$

$A_t = [(B\ x\ T) - n\ (d\ x\ t)]$

$A_c = (d\ x\ t)$

$HC = F\ x \cos\theta$

$R = \sqrt{VC^2 + HC^2}$

$VC = F\ x \sin\theta$

$D = \frac{Mass}{Volume}$

$W = mass\ x\ ga$

$Stress = \frac{Load}{Area}$

$Z = \frac{m}{f}$

$X = \frac{\sum Ax}{\sum A}$

$I_{yy} = \frac{db^3}{12}$

$I_{xx} = \frac{\pi D^4}{64}$

$I_{xx} = \frac{BD^3}{12} - \frac{bd^3}{12}$

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ADDENDUM 1

EXAMINATION NUMBER: | | | | | | | | | | | | | |

QUESTION 5

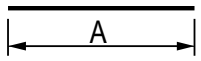
1	2	3	4	5	6	7	8	9	10	11	12
Member	Reinforcement					Bending dimensions for shape codes (SANS 282)					
Mark; size & No. off	Mark	Type and size	No. in each	Total No.	Length (mm)	Shape code	A	B	C	D	E or R
Rectangular Beam and Column **2 off**	1										
	2										
	3										
	4										
	5										
	6										
	7										

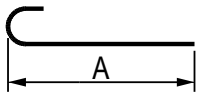
TABLE

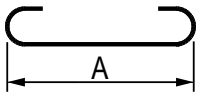
Copyright reserved

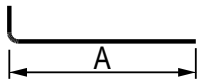
ADDENDUM 2

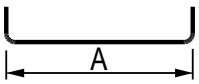
SANS 282 SUMMARY OF SHAPE CODES

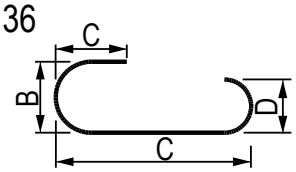
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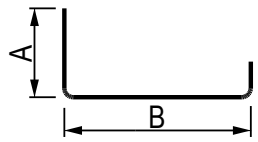
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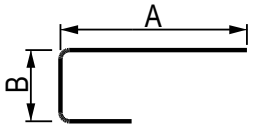
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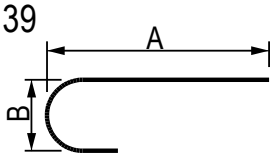
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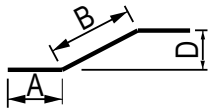
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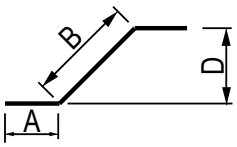

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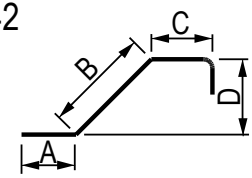


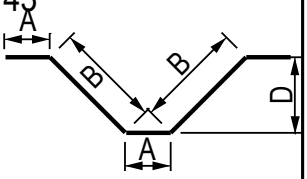
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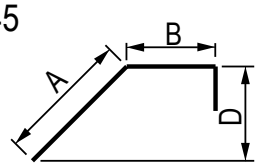
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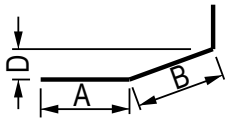
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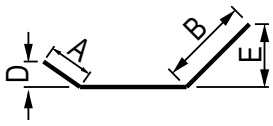


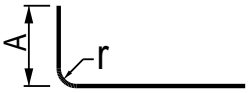
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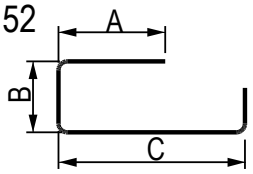
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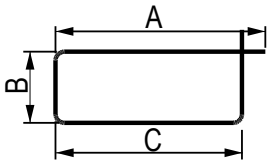
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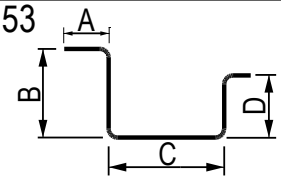
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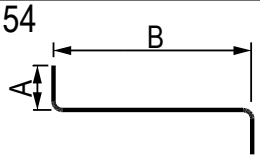
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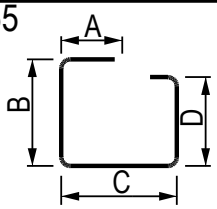
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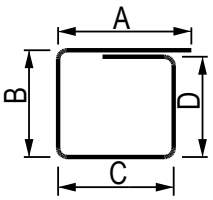
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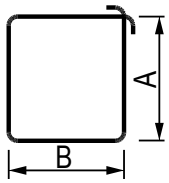


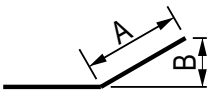
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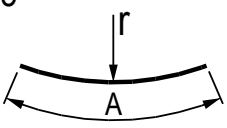
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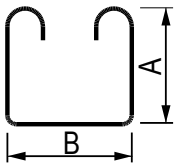
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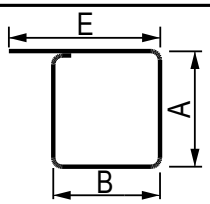


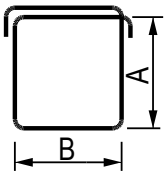
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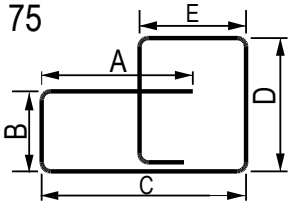
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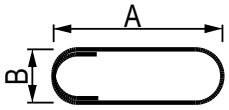
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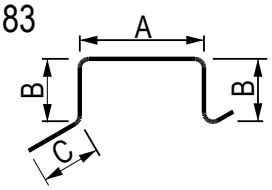
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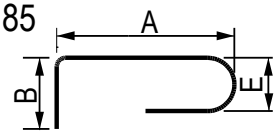
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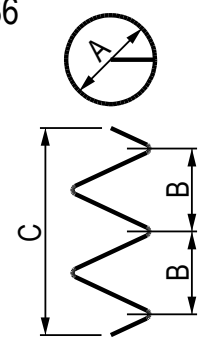
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